Prompt Sensitivity and Silent Action Drift in OpenVLA:Two-Stage Study on BridgeData V2

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Abstract

Vision-language-action (VLA) models promise a simple interface for robot control: the model sees an image, reads a language instruction, and predicts the next robot action. Recent work suggests that current VLA systems do not always use language as reliably as their task success numbers imply. This paper studies that problem in OpenVLA-7B with a fully reproducible local evaluation pipeline. Two complementary experiments are run on BridgeData V2 images. **Manual40** contains 40 hand-selected trajectories with manually written task prompts. **Auto50** contains 50 automatically sampled trajectories with prompts derived from run-family metadata. For every image, four prompt families are evaluated: normal, paraphrased, contradictory, and neutral. The predicted actions are compared using the first six action dimensions, measuring both L_2 drift and cosine similarity relative to the normal prompt. Across both experiments, paraphrases preserve behavior best, while contradictory and neutral prompts induce larger drift. Final-state images are more fragile than initial-state images in both branches. The paper does not claim a full safety solution. Instead, it offers a careful empirical slice of prompt robustness, a working local pipeline, and a concrete argument for using action drift as a future inference-time safety signal.

1 Introduction

Large vision-language-action models promise a simple idea: show the robot a camera image, tell it what to do in words, and let it predict the next robot action. OpenVLA is one of the most visible open-source examples of this idea and has become attractive because it can be run, adapted, and inspected without depending on a closed industrial stack [1]. At the same time, recent work has started to question how much these models truly depend on language once the scene is visually informative. LIBERO-X shows that robustness can drop sharply under structured perturbations [6]. LangGap argues that current VLAs often ignore language unless the benchmark forces semantic diversity by design [7]. IGAR shows that even contradictory instructions may still produce visually plausible actions, a failure mode described as linguistic blindness [8]. BayesianVLA frames part of the same behavior as *information collapse*, where the instruction becomes too predictable from the image and therefore stops influencing the action strongly enough [9].

This paper studies that problem from a practical and reproducible perspective. Instead of building a new benchmark from scratch, it takes an existing open model, OpenVLA-7B, and tests its sensitivity to four prompt families on BridgeData V2 images [3]. Two complementary evaluation branches are used. The first branch, **Manual40**, contains 40 hand-selected trajectories with task-specific language written manually from the scene. The second branch, **Auto50**, contains 50 automatically sampled trajectories from the scripted BridgeData split and uses language prompts derived from run-family metadata. For each image, a normal instruction is compared against a paraphrased version, a contradictory version, and a neutral version. Action drift relative to the normal prompt is then measured using the first six action dimensions, and initial and final states are analyzed separately.

The contribution is modest but concrete. First, the paper provides a local, reproducible OpenVLA evaluation pipeline that runs under WSL with 4-bit loading and a patch for the image–text attention-mask mismatch encountered during inference. Second, it reports a two-stage prompt-sensitivity study over 180 image evaluations in total: 80 images from Manual40 and 100 images from Auto50. Third, it shows a consistent pattern across

both branches: paraphrases preserve behavior best, contradictory and neutral prompts induce larger drift, and final-state images are more fragile than initial-state images. The paper does not claim to solve language robustness in VLAs. Instead, it provides a careful empirical slice of the problem, a concrete engineering recipe for reproducing the study, and a clear agenda for the next step: turning action drift into an inference-time warning signal.

2 Why This Matters in Practice

Language-conditioned robots will not receive one perfectly written benchmark instruction forever. A home robot may hear "put the pot on the towel," "place the pan on the cloth," or something vague like "deal with that pot." A stable policy should not change behavior dramatically just because the surface wording changed. The same issue matters in factories, hospitals, assistive settings, and education. When a policy is brittle to wording, the result is not merely an academic artifact; it becomes a deployment risk.

This is exactly why prompt robustness matters now. The recent VLA survey emphasizes generalization, multimodal alignment, system scalability, and safe deployment as open challenges for embodied intelligence [5]. SAFE and FailSafe show that the community is already moving toward detection and recovery, which means that action drift is no longer just a descriptive curiosity; it is part of the broader safety problem for deployable VLAs [10, 11]. The present paper contributes an earlier layer of that stack: a careful measurement of how much the action already changes when only the words change.

3 Background and Model Choice

A vision-language-action model is a model that takes vision and language as input and predicts an action for a robot. The "vision" part means images or video from a camera. The "language" part means an instruction such as "pick up the rigid object." The "action" part means the next motion command for the robot arm or end effector. Recent surveys describe this as the unification of perception, language understanding, and embodied action inside one framework [5].

OpenVLA was a natural choice for three reasons. First, it is open-source and widely used in the research community [1]. Second, it was designed to be adaptable and inspectable, which matters if the goal is to analyze behavior rather than merely report a benchmark score. Third, it can be run locally with a carefully managed setup, which made it realistic for a student-led project without depending on a large cluster. In the official OpenVLA paper, the model is described as a 7B-parameter open-source VLA trained on 970k real-world robot demonstrations [1].

Table 1: Why OpenVLA was used now, and where π_0 fits next.

Dimension	OpenVLA in this project	π_0 in future work
Openness and access	Fully open paper, code, and checkpoints	Public paper and growing ecosystem
Architecture style	Autoregressive token-style VLA	Flow-matching VLA
Local debugging value	High for this project	Promising, but not yet integrated here
Completed experiments	Yes	No

π_0 is also highly relevant. It is built around a flow-matching action model on top of a pre-trained vision-language model and was designed for general robot control, including dexterous and longer-horizon tasks [2]. In many ways, π_0 is a stronger stress-test candidate for a follow-up paper because it represents a different action architecture than OpenVLA. However, OpenVLA was chosen as the primary experimental model here because it was easier to instrument, easier to patch locally, and easier to run end-to-end inside the local WSL workflow. That is not a claim that OpenVLA is categorically better than π_0; it is a claim about experimental practicality and reproducibility for this project.

4 Research Gap

The cleanest research gap for the completed work is not "this paper solves silent failure." That would be an overclaim. The gap filled here is more careful: there was no systematic, locally reproducible measurement in this project pipeline of how much OpenVLA action vectors move when prompt wording changes across both

manually curated and automatically sampled BridgeData image subsets. In other words, before building a detector or a recovery system, a trustworthy map of prompt sensitivity was needed.

This gap is still worth filling because the broader literature mostly asks larger questions: do VLAs ignore language [7]? Can train-free attention recalibration repair some contradictions [8]? Can a trained detector identify failures across tasks [10]? Can a recovery model learn to reason about failures and repair them [11]? Those are important questions, but they do not remove the need for a concrete, low-level baseline showing how large the action changes already are in a real open-model workflow.

The contribution of the present paper is therefore an empirical bridge between broad failure theory and actual local model behavior. It answers practical questions a researcher or student might ask before attempting a safety layer: Are paraphrases mostly harmless? Are contradictory prompts much worse? Are neutral prompts also dangerous because they wash out task information? Are late or near-completion scenes more fragile than early scenes? The results show that the answer is yes to all four, with final-state scenes showing especially elevated drift.

5 Experimental Setup

Model and runtime. All experiments use OpenVLA-7B in 4-bit mode on a local WSL environment backed by an NVIDIA RTX 2060 GPU. The base model is loaded through the Hugging Face OpenVLA checkpoint with remote code enabled. Standard processor utilities and the model's built-in `predict_action` interface are used for inference.

Inference patch. During initial testing, a critical mismatch appeared between the text attention mask and the image-augmented embedding sequence. The practical symptom was a runtime failure after visual tokens were fused into the transformer input. The fix was to prepend a 256-token image attention mask before calling `predict_action`. This patch does not alter the learned OpenVLA weights; it only repairs the alignment expected by the model's inference path. This engineering step matters because, without it, the study would not be reproducible on the target local hardware.

Data and the 30 GB download. The first evaluation branch, Manual40, was built manually from 40 BridgeData V2 trajectories. For each trajectory, one initial frame and one final frame were saved, resulting in 80 total images. One task-specific natural-language command was written per trajectory, and three additional prompt variants were derived from it. The second branch, Auto50, was created automatically from the downloaded scripted BridgeData split. Fifty trajectories were sampled, one initial frame and one final frame were extracted for each, and task prompts were generated from the recorded run-family metadata. This produced 100 total images. The combined study therefore covers 180 image evaluations.

The full scripted subset is large enough to feel inconvenient on a student machine, which is exactly why it matters in a real project story. The 30 GB local download was not done for convenience; it was done to move from a toy pilot to a proper sample grounded in an official dataset. Once the data was local, it became possible to build an automated 50-trajectory branch with reproducible naming, metadata, and prompt generation. That is a major quality jump over a handful of random screenshots.

Prompt families. Each image is evaluated with four prompt types: *normal*, which states the intended task directly; *paraphrased*, which keeps the meaning but changes the wording; *contradictory*, which intentionally conflicts with the original task; and *neutral*, which removes task-specific information and asks only what action the robot should take. The central question is whether the action predicted by OpenVLA remains stable under meaning-preserving changes and how strongly it shifts under meaning-breaking or underspecified prompts.

Metrics. Only the first six action dimensions are analyzed, excluding the final scalar that often collapsed to a near-binary value in some runs. For each non-normal prompt, the following quantities are computed relative to the normal-prompt action $\mathbf{a}^{(n)}$:

$$d_{L_2}(\mathbf{a}^{(n)}, \mathbf{a}^{(p)}) = \left\| \mathbf{a}^{(n)} - \mathbf{a}^{(p)} \right\|_2, \tag{1}$$

where $\mathbf{a}^{(p)}$ is the action predicted for a perturbed prompt, and

$$\cos(\mathbf{a}^{(n)}, \mathbf{a}^{(p)}) = \frac{\mathbf{a}^{(n)} \cdot \mathbf{a}^{(p)}}{\|\mathbf{a}^{(n)}\|_2 \, \|\mathbf{a}^{(p)}\|_2}. \tag{2}$$

Lower L_2 means less drift; higher cosine means stronger directional agreement. Simple robustness counts are

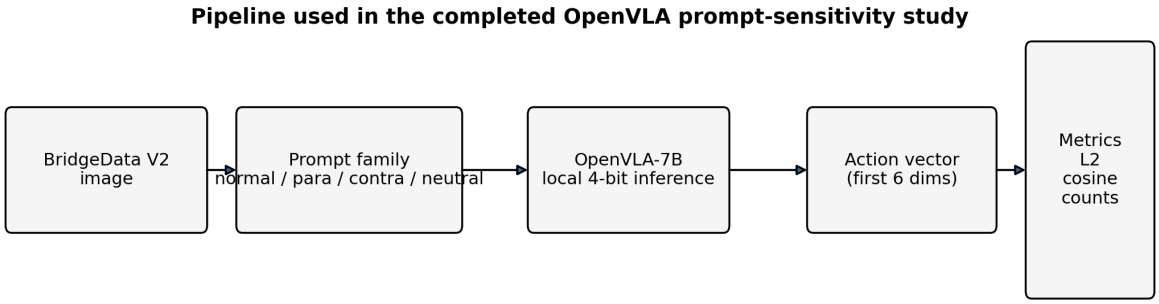

Figure 1: Pipeline used in the completed OpenVLA prompt-sensitivity study. A BridgeData V2 image is paired with four prompt families, passed through local OpenVLA-7B inference, and compared through action-space metrics relative to the normal prompt.

also computed: exact-all-same cases, near-all-same cases with maximum drift below 0.01, the number of times the paraphrase is the closest prompt to the normal action, and the number of times contradictory or neutral prompts are the farthest.

Analysis outputs. The study summarizes group means, simple robustness counts, a ranked list of high-drift cases, and confidence-interval plots derived from the completed results package. The present paper deliberately stays diagnosis-first: no simulator rollouts, no success-rate labels, and no active safety intervention are added here.

6 Results

Table 2 is the main quantitative summary. Lower L_2 means less drift. Higher cosine means more directional agreement with the normal prompt. The most important pattern is simple: paraphrases are always the closest prompt family, and final states are always harder than initial states.

6.1 Robustness counts

The count-based summary in Table 3 adds an important nuance. Manual40 still contains a few exact-all-same and near-all-same cases, especially on initial frames. Auto50 almost eliminates this effect. That suggests the manual subset contains some visually overdetermined scenes, but the broader sampled branch reveals more uneven language sensitivity.

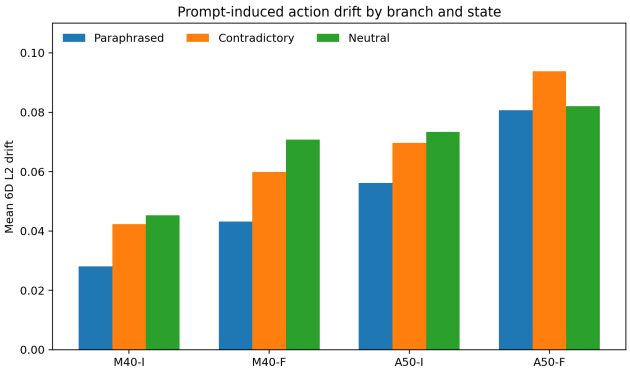

Figure 2: Mean six-dimensional L_2 drift from the normal prompt. Final scenes are more fragile than initial scenes, and paraphrases are the least disruptive prompt type in every group.

6.2 Representative high-drift cases

Table 4 lists representative high-drift cases from the completed results package. The strongest failures are not random: they cluster in final scenes and in visually crowded pick-and-place families.

7 Discussion

The results are consistent enough to support three claims. First, paraphrases are the most stable prompt family in every group studied. Second, contradictory and neutral prompts create materially larger drift than paraphrases, although the ordering between contradictory and neutral depends on the branch and state. Third, final-state im-

Table 2: Main quantitative summary. L2-P/C/N denote mean six-dimensional L_2 drift under paraphrased, contradictory, and neutral prompts. Cos-P/C/N denote the corresponding cosine similarities relative to the normal prompt.

Experiment	Group	n	L2-P	L2-C	L2-N	Cos-P	Cos-C	Cos-N
Manual40	Initial	40	0.028	0.042	0.045	0.679	0.586	0.426
Manual40	Final	40	0.043	0.060	0.071	0.480	0.219	-0.002
Auto50	Initial	50	0.056	0.070	0.073	0.484	0.269	0.171
Auto50	Final	50	0.081	0.094	0.082	0.387	0.157	0.220

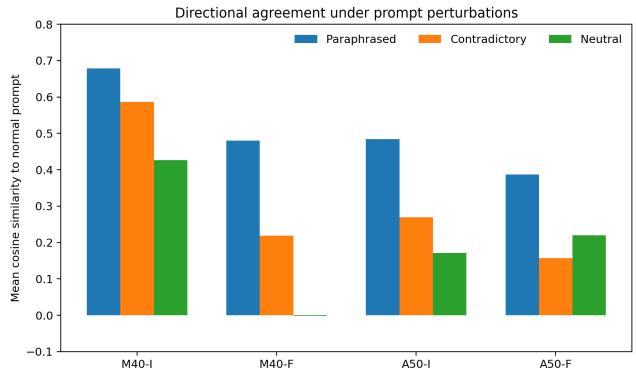

Figure 3: Mean cosine similarity to the normal prompt. The directional agreement analysis tells the same story as the L_2 chart from a different angle.

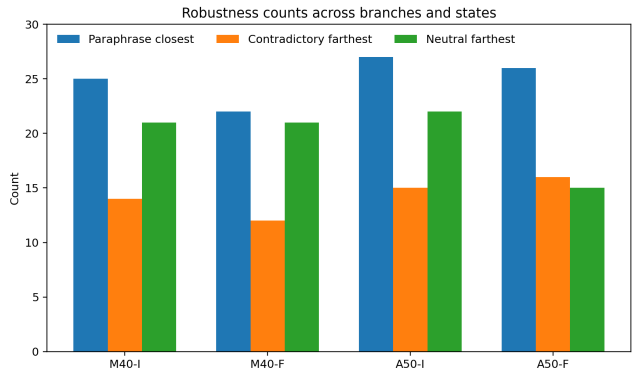

Figure 4: Robustness-count view of the same pattern. Paraphrases are usually closest to the normal action, while contradictory and neutral prompts are frequently the farthest.

Table 3: Robustness counts. Near-all-same means the maximum six-dimensional drift across the three perturbed prompts remained below 0.01.

Group	Exact	Near	Para close	Contra far	Neutral far
M40-I	1	8	25	14	21
M40-F	0	4	22	12	21
A50-I	0	1	27	15	22
A50-F	0	1	26	16	15

ages are more fragile than initial-state images in both Manual40 and Auto50.

In Manual40, the average six-dimensional L_2 drift rises from 0.028 under paraphrases on initial frames to 0.043 on final frames. Contradictory drift rises from 0.042 to 0.060, and neutral drift rises from 0.045 to 0.071. This branch therefore shows a clean monotonic fragility pattern from initial to final states. The same story largely repeats in Auto50 but at a higher overall drift level. Auto50 initial-frame drift is 0.056 for paraphrases, 0.070 for contradictions, and 0.073 for neutral prompts. Auto50 final-frame drift increases further to 0.081 for paraphrases, 0.094 for contradictions, and 0.082 for neutral prompts.

The cosine results tell the same story from a directional rather than magnitude-based perspective. In Manual40 initial frames, paraphrase cosine remains high at 0.679, while contradictory and neutral drop to 0.586 and 0.426. In Manual40 final frames the drop becomes sharper, especially for contradictory and neutral prompts, with the neutral mean crossing slightly below zero. Auto50 cosine values are lower overall, again suggesting that the broader sampled branch is harder. Yet even there, paraphrases preserve the strongest alignment in both initial and final groups.

The invariance counts add an important nuance. In Manual40, one exact-all-same case and several near-all-same cases still appear, especially on initial frames. Auto50 almost eliminates this effect. This suggests that the manual subset contains some images where the task is visually overdetermined, but once the sampled branch becomes broader, OpenVLA is less likely to collapse to identical outputs across all prompts. In other words, the model is not uniformly blind to language; it is inconsistently sensitive, which may be even more dangerous in practice because the failure mode is scene-dependent rather than constant.

Finally, the top-failure ranking shows that the largest

Table 4: Representative high-drift cases. Values are six-dimensional L_2 distances from the normal prompt.

Experiment	Group	Task / Image	L2-P	L2-C	L2-N
Auto50	Final	`traj015_final.jpg`: pick up the soft toy	0.362	0.166	0.189
Auto50	Final	`traj004_final.jpg`: pick up the object and place it in the bin	0.297	0.253	0.100
Manual40	Final	`traj012_final.jpg`: pick up the paper	0.092	0.239	0.294
Auto50	Initial	`traj024_initial.jpg`: pick up the object	0.144	0.285	0.225
Auto50	Final	`traj049_final.jpg`: pick up the utensil	0.034	0.278	0.048
Auto50	Initial	`traj006_initial.jpg`: pick up the object	0.059	0.245	0.129
Auto50	Initial	`traj001_initial.jpg`: pick up the rigid object	0.105	0.241	0.181
Manual40	Final	`traj019_final.jpg`: put the green object into the drawer	0.068	0.240	0.073

drifts concentrate in final scenes and in pick-and-place families involving soft toys, bins, and crowded object layouts. This pattern is intuitive. Final scenes often look more ambiguous because the goal may already be partly achieved, objects may be occluded, and the action space may contain both completion and correction motions. Under these conditions, a language change can more easily flip the model from a stable continuation policy to a qualitatively different adjustment policy. This observation points toward a useful next step: drift itself may be an early warning signal for prompt confusion.

8 Where the Project Succeeded and Stayed Incomplete

This project succeeded in several ways that matter for real research. It established a working local OpenVLA pipeline on modest hardware, fixed a real inference bug, completed two complementary experiment branches rather than stopping at a tiny pilot, produced clear and stable numerical patterns rather than ambiguous noise, and generated reusable metadata, CSV results, summary tables, and failure-case lists. In other words, it built the measurement layer that a stronger future silent-failure paper would require before claiming a safety benefit.

At the same time, the project did not yet close the loop with a simulator outcome metric such as task success or collision rate. It did not run π_0 or another second VLA as a cross-model ablation. It did not test an intervention such as re-prompting, abstention, or active visual recovery. The seventh action dimension was not used in the main analysis because it did not behave helpfully enough for the chosen metrics. The Auto50 prompts were derived from run-family metadata, which is reproducible but semantically weaker than careful hand writing. The project also used static images, not full rollout dynamics, so it measures action drift rather than full embodied success.

9 Practical Applications

The immediate practical value of this work is not that it already makes a robot safer. The value is that it shows where safety work should start. In home robotics, instructions may come from children, elders, or non-native speakers, which means wording variability is unavoidable. In factory settings, language-conditioned interfaces may be compressed, paraphrased, or context-dependent. In assistive robotics, especially near task completion, a system that silently follows visual priors while underusing language could become misleading or unsafe. The present study provides a quantitative basis for deciding when a prompt perturbation is mild and when it is risky enough to justify a fallback action.

There is also educational value. Because the full pipeline runs locally under WSL on modest hardware, the project provides a repeatable path for students and small labs to study VLA robustness without needing large robot fleets or a major cluster. That is useful beyond this single paper: it lowers the barrier to entry for future diagnostic, benchmark, and safety work.

10 Limitations and Future Work

This study has four important limitations. First, it evaluates action-space drift rather than true task success. The paper shows that the predicted action changes, but it cannot yet say how often that change would produce an execution failure, a recoverable deviation, or a harmless alternative trajectory. A simulator study or replay-based

evaluation is needed for that step. Second, only one base model, OpenVLA-7B, is evaluated rather than a cross-model panel. This makes the paper cleaner and more reproducible, but it limits claims about whether the same pattern holds across other VLAs such as π_0. Third, the Manual40 and Auto50 branches are not identical in construction: one uses task-specific manual prompts, while the other derives language from run-family metadata. This is a strength for breadth but a limitation for strict apples-to-apples comparison. Fourth, an inference-time attention-mask issue had to be patched to make the model run locally. Although this does not alter the learned weights, any local engineering patch raises the question of whether hidden implementation details could affect robustness.

These limitations also explain why the problem is not already solved. Prompt robustness in VLAs is difficult for at least three reasons. The first is data bias: in many robot datasets, the scene already predicts the task strongly enough that the model can succeed by following visual shortcuts, a problem emphasized by LangGap and BayesianVLA [7, 9]. The second is evaluation cost: measuring language sensitivity with real robots or large simulation suites is expensive, and fine-grained prompt perturbations multiply that cost quickly. The third is systems complexity: once a model mixes images, text, and low-level control, failures can originate from the data, the fusion architecture, the action head, or the runtime implementation itself.

Future work should move in three directions. The first is *outcome-aware evaluation*: run the highest-drift cases in simulation and test whether drift predicts task failure. The second is *inference-time safety*: treat drift as a detector that can trigger re-prompting, abstention, or a safe retreat behavior, similar in spirit to the failure-detection literature represented by SAFE and FailSafe [10, 11]. The third is *stronger causal testing*: extend the study across multiple VLA families and across structured robustness benchmarks such as LIBERO-X and LangGap, which were designed to expose language and scene-understanding weaknesses more systematically [6, 7].

11 Conclusion

This paper presented a fully reproducible prompt-sensitivity study for OpenVLA-7B using two BridgeData V2-based evaluation branches. Manual40 provides a hand-curated, task-specific view; Auto50 provides a broader sampled validation view. Across both branches, paraphrases preserve action behavior best, while contradictory and neutral prompts produce larger drift. Final-state images are consistently more fragile than initial-state images. The results do not prove that OpenVLA ignores language everywhere, but they do show that language sensitivity is uneven, scene-dependent, and weak enough to matter. A robot policy does not need to fail on every contradictory prompt to be unsafe; it only needs to fail unpredictably. The present study offers a concrete local pipeline, a set of reproducible results, and a practical basis for the next step: turning action drift into a real-time warning signal for safer VLA deployment.

A Exact numbers used in the paper

Manual40 initial drift: 0.028126 / 0.042365 / 0.045330
Manual40 final drift: 0.043141 / 0.059920 / 0.070772
Auto50 initial drift: 0.056164 / 0.069716 / 0.073405
Auto50 final drift: 0.080668 / 0.093857 / 0.082103

The ordering is stable enough to support the main qualitative claim: paraphrases preserve behavior best, while contradictory and neutral prompts produce larger action drift. The Auto50 final group is the only case where contradictory prompts exceed neutral prompts by a visible margin.

B What is stopping this from being fully solved today?

First, the training data often lets the image predict the task too well, so the language channel is weak. Second, evaluation is expensive because each prompt variant multiplies the number of rollouts needed for trustworthy safety claims. Third, the full stack is complicated: data bias, model fusion, action decoding, and runtime engineering can all create or hide the problem. Fourth, even when a failure is detected, deciding what the robot should do next is not trivial; a safe fallback may reduce harm, but it may also reduce task completion if used too often. These are exactly the reasons why future work should focus on outcome-aware evaluation and inference-time safety mechanisms, not just more descriptive benchmarks.

C Extended engineering notes

The present study was carried out in a resource-constrained but realistic setting: Ubuntu under WSL2, a Python virtual environment, consumer-grade GPU hardware, and iterative debugging of the inference stack. That engineering path matters because many student projects fail not at the idea stage but at the reproducibility stage. A useful robotics paper does not need to begin with a giant robot fleet. It can begin with a local, open model, a citable dataset, careful perturbation design, and honest logging of what breaks and how it is repaired.

The most important engineering failure was the attention-mask mismatch after visual token fusion. Secondary problems included environment setup friction, missing dependencies, real-image path failures, and the need to maintain a stable 4-bit inference environment. Recording those details is not merely autobiographical. It turns the paper into a practical starting point for the next person who wants to reproduce the result.

D Why OpenVLA was the right first model

OpenVLA was not chosen because it is universally better than all alternatives. It was chosen because it made the first complete result experimentally feasible. The model is open, visible in the literature, compatible with a growing ecosystem, and inspectable enough to patch locally [1]. That matters for a diagnosis-first study. By contrast, π_0 is the correct next model for a follow-up paper because it would test whether the same prompt-drift pattern survives across architectures [2]. If both models show the same ordering, the result becomes more likely to reflect a broader property of present-day VLAs. If the ordering changes, that difference itself becomes scientifically useful.

E Plain-English glossary

Action vector: a list of numbers that tells the robot how to move next.
BridgeData V2: a large robot-learning dataset with many manipulation trajectories.
Contradictory prompt: a sentence that conflicts with the intended task.
Cosine similarity: a number that tells whether two vectors still point in roughly the same direction.
Final-state image: an image taken near the end of a task, when the next move may need more precise language grounding.
Information collapse: a setting where the language signal becomes too weak compared with the visual signal.
Initial-state image: an image taken near the beginning of a task.
L_2 **distance:** a number that tells how far apart two vectors are.
Neutral prompt: a generic instruction that removes task-specific meaning.
Paraphrased prompt: a sentence that keeps the same meaning but changes the wording.
Prompt-induced action drift: the change in predicted robot action when the wording changes but the image stays the same.
WSL2: Windows Subsystem for Linux, the local environment used to run the pipeline.

F Beginner reproduction checklist

A beginner-friendly way to think about the full project is simple. First, choose one open robot brain that can actually be run. In this project, that model was OpenVLA. Second, choose one real dataset that can be cited. In this project, that dataset was BridgeData V2. Third, create a small but serious subset rather than a fake toy demo. Fourth, test multiple prompt types on the same image. Fifth, measure how much the action changes. Sixth, interpret the result honestly. That is the full project skeleton.

The local execution path was: install the Python environment, verify that the GPU was visible, load OpenVLA in 4-bit mode, fix the attention-mask bug, run a dummy-image test, run a real-image test, build metadata for Manual40, build metadata for Auto50, run the four prompt families, summarize the results, and then write the paper around what actually happened rather than around an imagined perfect outcome.

A beginner-friendly way to think about the metrics is also simple. L_2 distance asks, "How far apart are the two action vectors?" Cosine similarity asks, "Are they still pointing in roughly the same direction?" If paraphrases stay close while contradictions and neutral prompts move away, then the model has at least some useful language sensitivity. If all prompts produce almost the same action, the model may be ignoring language. If all prompts produce chaotic changes, the model may be fragile in a different and more obvious way.

G Extended practical implications

This project is about checking whether a robot brain stays calm when the words change. The same picture is kept, only the sentence is changed, and then the next movement is measured. That sounds small, but in robotics a wording change can become a motion change. That connection is exactly why the problem matters. If we want trustworthy robots, we cannot only ask whether they can do the task in a benchmark. We also have to ask whether they still do the right thing when people speak in slightly different ways.

One future safety pipeline suggested by the results is straightforward. The model could monitor action drift, attention drift, or another lightweight internal score at inference time. If that score rises sharply, the policy could ask for a clearer instruction, stop for confirmation, or take a safe re-observe action. The present paper does not claim that system has been built, but it provides the measurement layer and the high-risk prompt families that such a system would need.

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